TOPIC SEVEN - VENTILATION SYSTEMS

General Introduction.

Ventilation is the process of replacing stale or contaminated indoor air with clean outdoor air. As buildings become more airtight to save energy, an unintended consequence is an accumulation of internally generated contaminants that cause deficient indoor air quality (IAQ). Deficient IAQ is a serious problem in all buildings since it negatively impacts indoor occupants' health, cognitive function, productivity and wellbeing. Poor ventilation can lead to indoor air pollution, which affects health, comfort, and efficiency.

Indoor air in residences can be quite unhealthy.
In fact, the U.S. Environmental Protection Agency (EPA) states that:

- The average person receives 72% of their chemical exposure at home.
- Indoor levels of pollutants may be two to five times – and occasionally more than 100 times higher than outdoor levels.
- High amounts of indoor air pollutants are of particular concern because most people spend about 90% of their time indoors.
- Indoor air pollution is a top-five environmental risk to public health.

Common indoor pollutants in Kenyan buildings include:

- Smoke from cooking fuels (charcoal, kerosene, LPG).
- Dust and vehicle exhaust (especially in urban Nairobi, Mombasa, and Kisumu).
- Volatile organic compounds (VOCs) from paints, furniture, and cleaning detergents.
- Carbon monoxide (from faulty appliances).
- Mold and dampness due to humidity and poor airflow.

Impact of poor ventilation:

- Respiratory issues, allergies, and asthma.
- Eye and skin irritation.
- Fatigue, loss of concentration, and overall discomfort.
- Long-term exposure may increase risk of serious diseases.

To enhance IAQ, there is need for assisted ventilation.

Reasons for implementing ventilation

Ventilation is necessary for the following reasons:

a) **Maintaining the quality of indoor air.** Effective ventilation helps reduce indoor air pollutants such as volatile organic compounds (VOCs), carbon monoxide, and particulate matter, which can have adverse effects on human health. For the building to perform as intended the occupants need to be able to inhale air free from odors, pollutants and allergens. This will also ensure the safety and health of the occupants.

b) **Regulating temperature-**Ventilation systems play a crucial role in regulating temperature within a building or enclosed space. These systems help maintain a comfortable and healthy indoor environment by controlling the flow of air, which in turn impacts the temperature within the space. By facilitating the exchange of indoor and outdoor air, ventilation systems can help regulate temperature in various ways. Due to this the internal temperature is maintained at a constant comfortable level.

c) **Controlling humidity levels-**Ventilation systems help remove humid air produced in activities like cooking, washing, and showering. The system prevents build-up of this moisture in the interior of the building. Dry air from the outside is driven into the building diluting the humid air. This continuous cycle ensures there isn't any moisture build up which can lead to various issues such as mould growth, dampness, and structural damage. This will also prevent condensation on windows, walls and other surfaces.

d) **Remove airborne pathogens**-Ventilation can help reduce the spread of airborne pathogens such as viruses and bacteria, which is particularly important in crowded or enclosed spaces to minimize the risk of airborne transmission of diseases. By continuously circulating and exchanging air, ventilation systems can reduce the concentration of airborne pathogens, thereby lowering the risk of transmission within enclosed spaces.

e) **Enhance occupant comfort-**Proper ventilation can improve indoor air circulation, reducing stuffiness and odours and creating a more comfortable and pleasant indoor environment for occupants. By effectively managing these elements, ventilation systems contribute significantly to creating a pleasant and productive indoor environment for building occupants.

f) **Air Quality-**Ventilation helps remove indoor air pollutants such as volatile organic compounds (VOCs), carbon dioxide, and particulate matter. It ensures a steady supply of fresh air, improving indoor air quality and promoting better health for occupants.

g) **Odor Removal** -Effective ventilation helps eliminate unpleasant odors from indoor spaces, whether they originate from cooking, cleaning chemicals, or other sources.

h) **Health and Comfort** -Adequate ventilation contributes to the overall comfort and well-being of building occupants. It reduces stuffiness, prevents the buildup of stagnant air, and promotes a sense of freshness.

i) **Safety-**Ventilation is essential for removing potentially harmful gases, such as carbon monoxide, and for diluting indoor air pollutants to safe levels. It is especially critical in areas where combustible gases may be present.

j) **Compliance-**Building codes and regulations often mandate minimum ventilation requirements to ensure the health, safety, and comfort of occupants. Compliance with these standards is necessary for legal and regulatory purposes.

Considerations When Choosing a Ventilation System

Building specific factors

- **Location-** Assessing the local air quality and climate at the design stage can help to determine which ventilation method will be the most effective. Natural ventilation systems are reliant on the external air quality being higher than inside the building. Where this is not the case, mechanical solutions with air filtration will need to be used to reduce the concentration of particulate matter entering the building. Prevailing weather conditions can also have an impact.
- **Geometry-** The volume, shape and organization of the internal space all affect the way air flows through a building. Computational fluid dynamics (CFD) software can be used to understand how air moves through a space and which ventilation methods will be the most effective. This is particularly important to understand when specifying a passive system which relies on natural wind and thermal buoyancy.
- **Construction-** The building fabric can also influence the ventilation requirements. All buildings will have some level of air infiltration, where internal and external air leaks in and out through gaps in the construction. Older buildings will have high infiltration rates, whilst modern constructions are built to achieve increasingly higher airtightness for thermal efficiency This will have an impact on the ventilation rate required, with higher levels needed the less natural air movement occurs. Solar heat

gains due to glazing, orientation or heavy building materials will also need to be taken into consideration as these will increase cooling demand during hotter months.

- **Type+ Use-** From small schools and offices, to large production halls, every building type will carry its own specific set of required ventilation rates, air hygiene levels, temperature parameters and may even be subject to specific legislation. Internal activities, occupancy levels and occupancy type will also impact the air quality and therefore what rate of ventilation is required. It is also important to establish how the ventilation system will be used; for example, will it be operated throughout the day during occupancy or used for night cooling strategies?
- **Materials+ finishes**- Related to both the construction and the building type or purpose are the materials used inside the building. Paints, carpets, flooring, furniture, and other furnishings can all emit molecules of what are known as volatile organic compounds (VOCs), which can be inhaled by occupiers, causing all kinds of health issues. It is important that these are also considered when calculating the air quality of a space and, therefore, the ventilation rate required.

Project requirements

- **Regulations + standards-** Buildings may be subject to different regulations depending on their type, occupancy or the ventilation methods chosen. Additionally, some projects may be striving to achieve voluntary building standards relating to sustainability or health and well-being, and must achieve defined levels of energy efficiency and indoor air quality (IAQ).
- **Energy efficiency-** In addition to ventilation performance, the impact of the chosen system on the building's overall energy demand should be considered. Natural ventilation methods are of course the most efficient option to keep demand low. However, in buildings where mechanical ventilation is required, specifying more energy efficient fans or creating a hybrid system, using both active and passive ventilation strategies, can help to minimise the load.
- **Budgetary constraints-** As with any aspect of a building project, budget is key. It is not only important to consider the capital costs of the system, but the costs it will incur during its lifetime and how these balance with the benefits it will bring. Whilst under-specifying can result in a ventilation system that is not fit for purpose, over-specification can result in unnecessary cost and, in the case of mechanical means, higher energy bills.

Operational requirements

- **Noise Level:** Consider the noise level produced by the ventilation system, especially in spaces where low noise is crucial. Some environments, like offices or bedrooms, may require quieter systems.
- **Aesthetics:** Some systems can be incorporated to sit seamlessly within a building's functional and aesthetic design, such as hidden automated drives in windows, and skylights, or mechanical systems encased in suspended ceilings. For mechanical systems, it is also important to determine both the interior room space needed for ductwork, and the exterior roof or wall space for the external fan units, and the impact these may have on the overall building aesthetics.

Types of ventilation mechanisms

- Natural ventilation;
- Mechanical extract - induced inlet;
- Mechanical input - forced extract;
- Mechanical inlet - mechanical extract

A. NATURAL VENTILATION

This is the usual method of ventilation in domestic dwellings and many small office buildings.

The term natural ventilation relates to the air flow in a building that is caused by three natural factors:

1. temperature differences (thermal density) between the inside and the outside of the building;
2. wind forces around the building;
3. a combination of (1) and (2)

Natural ventilation has the advantage that no power supply is required: hence there are no fans and maintenance costs. It has the disadvantage that, during the summer months, the temperature difference between the inside and outside is small, and with low wind forces will result in poor ventilation when it is most required. During the winter months, however, the temperature difference between indoors and outdoors is high, with corresponding high wind forces, resulting in high ventilation rates and high rates of heat loss.

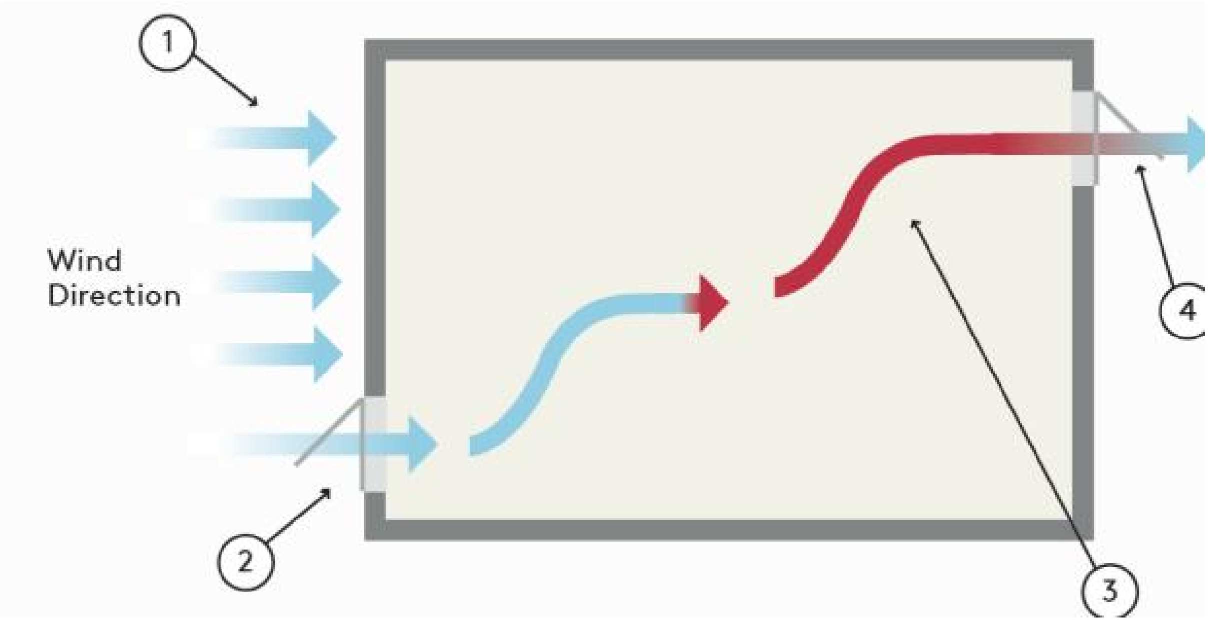
It is not possible to design a system according to a given natural ventilation rate, as this will vary with weather conditions. The cold outdoor air has a greater density, for a given volume, than the warmer indoor air. This density difference causes the cold dense air to enter through low-level openings, displacing the warmer indoor air and creating air changes. The displaced warmer air escapes through high-level opening.

Natural ventilation techniques:

a) Cross ventilation / wind force

Cross ventilation is a wind-driven process;

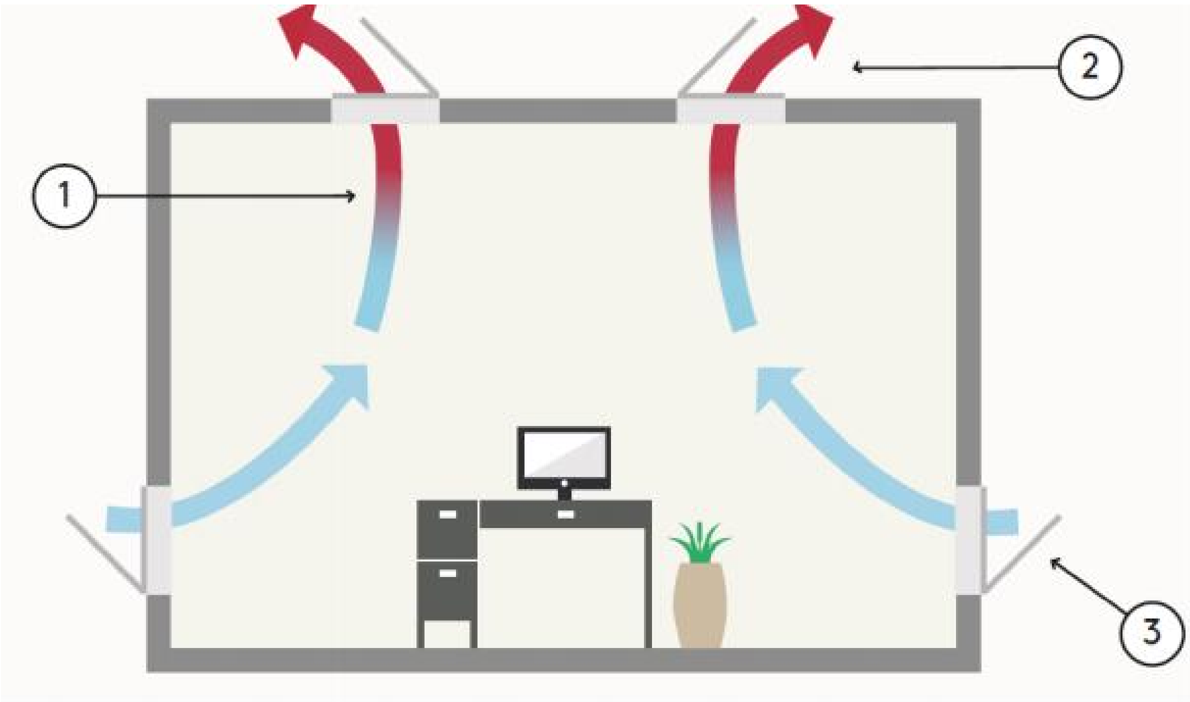
1. When rising temperatures or declining air quality triggers the system, or it is manually activated, vents or windows on either side of a space are opened.
2. Fresh air then flows into the building from the high-pressure side facing the wind and travels through the space.
3. The warm, stale air within the building is drawn out of the building by the negative air pressure difference caused by the prevailing wind direction, creating a continuous current through the space.
4. Typically, these air outlets are installed at a higher level to account for the buoyancy of warm air.



b) Stack ventilation / thermal convection

Stack ventilation is driven by thermal buoyancy

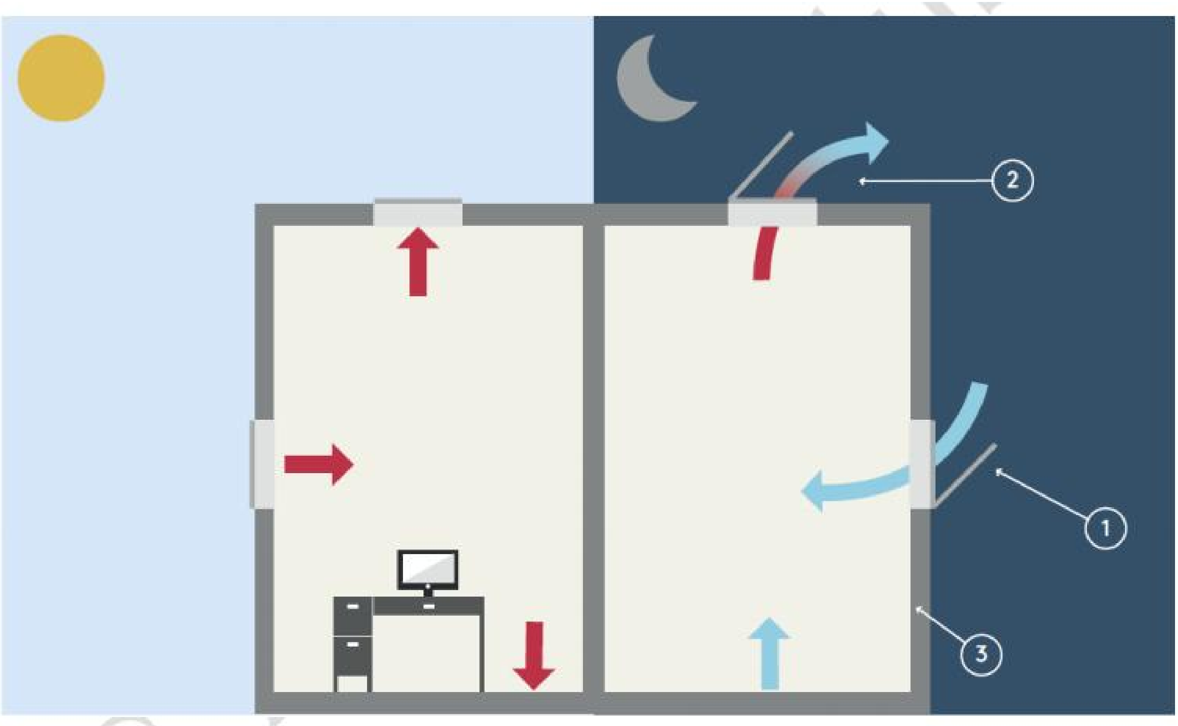
1. Warmer air is less dense than cooler air and so rises to the top of the space, creating an area of positive pressure.
2. When the system is activated and the roof and side vents or windows are opened, this is then drawn out of the building as the external air pressure will be lower than that inside the building.
3. Meanwhile, this creates negative pressure at the bottom of the space, drawing the fresh air from outside in through the low-level inlets



c) Night cooling techniques

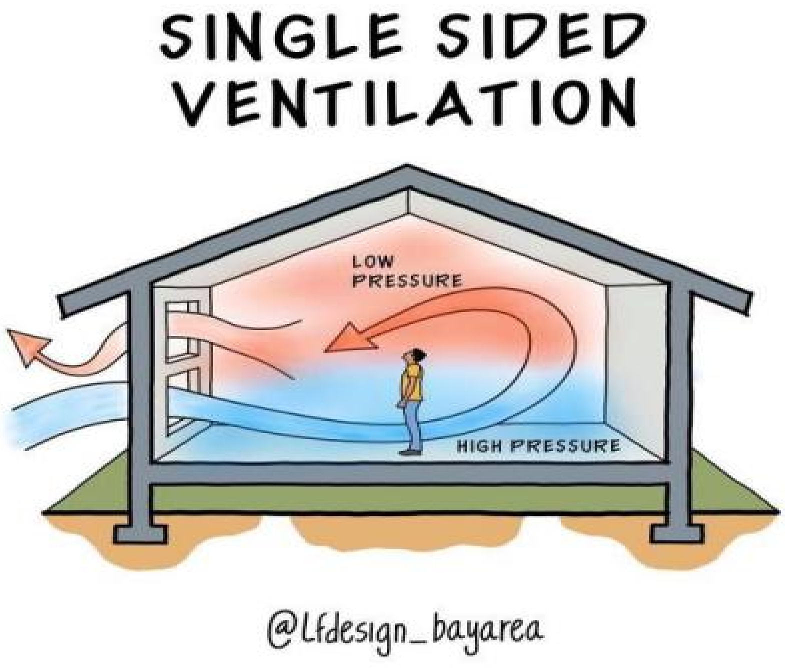
Sometimes called night flushing, night purge strategies capitalize on the natural temperature differences at night:

1. Automatically opening windows, skylights or vents are set to open for a period of time during the night when the building is unoccupied.
2. When the outside temperature is below the internal building temperature (as indicated by sensors), it removes the warm, stale air generated during the day by people, lighting and other equipment.
3. It also cools the mass of the building down, releasing any heat stored in the building fabric, such as floors and walls, ready for the next day



d) Single-sided ventilation

Single-sided ventilation involves the use of one opening to allow for air exchange between the indoor and outdoor environments. It is commonly used in buildings with narrow floor plans or where the building is located close to a busy road or in a noisy area. The opening may be a window, door, or vent, and it allows for the natural flow of air into and out of the building. Single-sided ventilation is useful in moderate climates with moderate wind speed



Natural ventilation strategies

Some natural ventilation strategies for a home include:

- **Building Orientation and Location:** Homes should be oriented so that the windward wall is perpendicular to the prevailing summer wind direction.
- **Building Form and Dimensions:** Naturally ventilated buildings should not be too deep to allow wind-driven air to pass through easily.
- **Window Typologies and Controls:** Different types of windows result in varying ventilation rates. External elements like trees or adjacent buildings may obstruct airflow
- **Air Inlets and Outlets:** Air inlets should be low in the room, while outlets should be higher and across the room from the lower inlet. Ridge vents can enhance air movement.
- **Encouraging Convective Air Movement:** Utilize convective ventilation, where warm air rises to escape through higher openings, drawing in cooler air from lower openings.
- **Automated Ventilation Systems:** Install systems that adjust window openings based on internal temperatures to regulate heat and airflow effectively

Merits of natural ventilation systems

- Reduces energy consumption and costs, particularly in mild climates.

- Offers lower maintenance expenses compared to mechanical systems.
- Enables better indoor air quality by diluting pollutants and controlling humidity
- Promotes a healthier living environment by removing contaminants and excess moisture.
- Allows for passive cooling and heating, enhancing thermal comfort.
- Requires little additional space for installation, unlike mechanical systems.
- Supports sustainability efforts since it generates zero greenhouse gas emissions.
- Can serve as an alternative to air-conditioning units, conserving energy resources
- May be more acceptable to residents who prefer natural solutions over mechanical systems.

However, natural ventilation systems are dependent on external factors such as climate and building design, and they may not be optimal in extreme environments. Additionally, natural ventilation requires proper planning and design to maximize its potential.

Demerits of Natural Ventilation systems

- **Dependence on weather:** Natural ventilation relies on wind and temperature differences making it susceptible to changes in atmospheric conditions.
- **Limited cooling capacity:** Natural ventilation may not be sufficient for cooling during periods of still air or high humidity.
- **Variable airflow patterns:** Natural ventilation can lead to uneven distribution of air throughout a building, resulting in uncomfortable areas.
- **Seasonality:** Natural ventilation may not be effective year-round, requiring additional cooling or heating systems depending on the climate.
- **Pollution prevention:** Natural ventilation does not filter out pollutants, relying solely on dilution to improve indoor air quality.
- **Regulatory constraints:** Local regulations may limit the use of natural ventilation, particularly in buildings with strict safety requirements.
- **Occupant control:** Natural ventilation may not provide users with direct control over airflow, which could lead to discomfort.

B. MECHANICAL EXTRACT, INDUCED INLET

The second method of ventilating a building uses an extractor fan, creating a negative pressure within the space. With this method a set flow rate can be achieved, as the fan will overcome the stack and wind effects: hence the system is not at the mercy of the weather.

This system causes the inside of the building to be held at a negative pressure, so air will be drawn in from outside or from surrounding spaces that are at a higher pressure.

A space may need to be held at negative pressure, as in lavatories, kitchens and process areas, for example. In such cases air will flow into the negative-pressure zone: hence odours or toxic gases will not escape to other areas. The fan, with its running and maintenance costs, makes this system more expensive than the natural method. Figure 5.3 shows a typical arrangement of this method of ventilation.

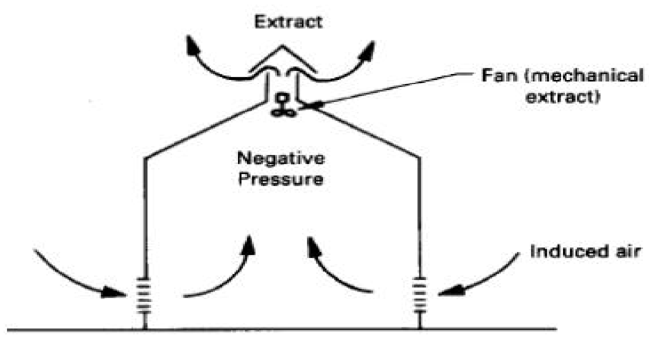


Figure 5.3 Mechanical extract, induced input

C. MECHANICAL INPUT, FORCED EXTRACT

If heating is provided, this is known as a plenum system. This is a ducted system, which may provide air to a space in one or more of the following conditions:

- untreated air (no filter);
- tempered air by means of a heater battery (heated to near room conditions);
- warm air at a temperature high enough to take care of the fabric and ventilation losses from the building.

With air being forced into the space the pressure is positive: hence all leakages are outwards. More sophisticated applications of a pressurized system, combined with a suitable extractor, are found in hospital operating theatres. The positive pressure produced by sterile air entering the room ensures that all leakages are outwards. This outward leakage ensures that contaminated air at a lower pressure in the surrounding rooms will not enter the space. Figure 5.4 shows a typical plenum system.

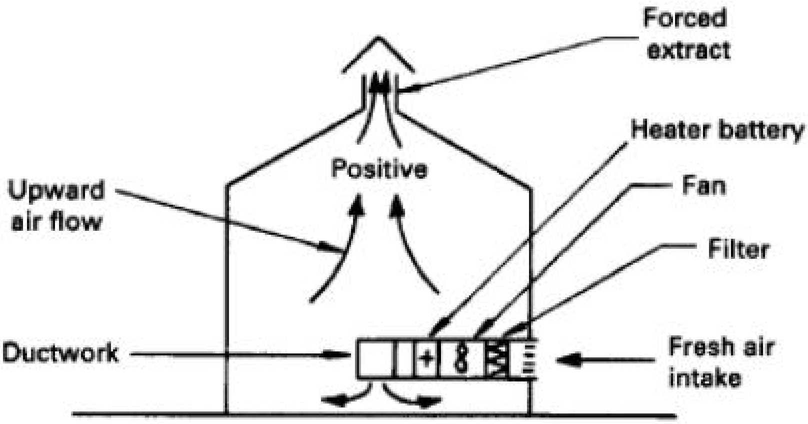


Figure 5.4 A typical plenum system (mechanical input, forced extract)

D. MECHANICAL INPUT, MECHANICAL EXTRACT

The final method of ventilation is that in which a powered extract and input system are provided. The nature of the pressure produced in this instance depends on the rate of extraction to input, or vice versa. It is no use if the input fans are handling the same amount of air as the extract fans, as this situation will result in a neutral condition, with the disadvantages of the natural system.

Figure 5.5 shows a typical layout of this type of system. From the foregoing statements the advantages and disadvantages of such a system are obvious

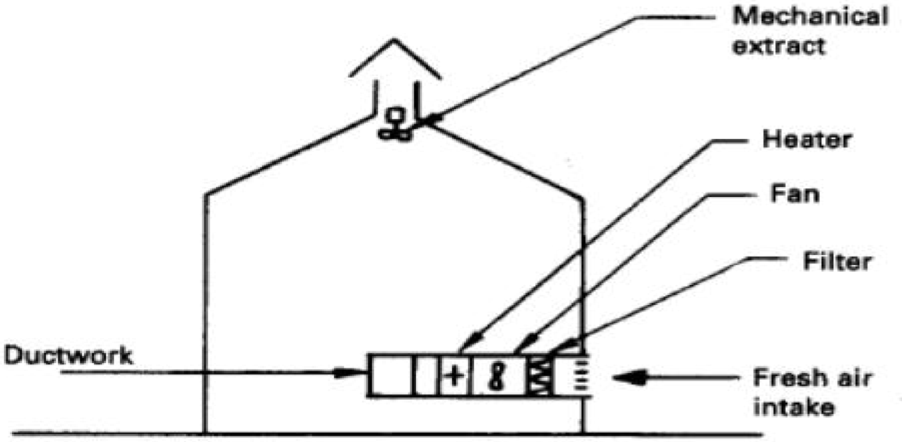


Figure 5.5 Mechanical input, mechanical extract

ADVANTAGES OF MECHANICAL VENTILATION SYSTEMS

- Consistent Airflow: Mechanical ventilation systems provide a consistent flow rate regardless of external factors like temperature and wind
- Controlled Airflow Direction: The direction of airflow can be controlled, ensuring efficient circulation and pollutant removal.
- Integration with Air Conditioning: These systems can easily integrate with air conditioning units, allowing for precise control over indoor temperature and humidity levels
- Filtration Capabilities: Mechanical ventilation systems can incorporate filtration systems to remove harmful microorganisms, particulates, gases, odors, and vapors from the air, enhancing indoor air quality.
- Energy Efficiency: Despite initial costs, these systems can be energy-efficient in the long run, contributing to cost savings and environmental benefits.

Mechanical ventilation systems offer a range of benefits such as improved indoor air quality, controlled airflow, and energy efficiency, making them essential for maintaining a healthy and comfortable indoor environment.

DISADVANTAGES OF MECHANICAL VENTILATION SYSTEMS

- Higher energy consumption: Compared to natural ventilation, mechanical systems consume more energy, primarily due to the use of electric motors and fans.
- Increased maintenance costs: Mechanical ventilation systems require routine maintenance to ensure optimal performance and air quality.
- Potential for increased heating and cooling costs: Introducing fresh air from outside can lead to increased demands on heating and cooling systems.
- Risk of spreading pollution: Improper installation or malfunctioning of mechanical ventilation systems can lead to the spread of pollutants, including fumes from appliances like water heaters and furnaces.
- Noise and vibration concerns: Fans and ductwork can generate noise and vibration, affecting the comfort of residents.
- Possible depressurization risks: Overzealous ventilation can lead to depressurization, causing back drafting of appliance emissions into the occupied spaces.
- Complex installation and design: Mechanical ventilation systems require careful planning and design to achieve optimal performance and energy efficiency